

# AMBIGUITY

Course: Automata Theory

Lecturer: Solita

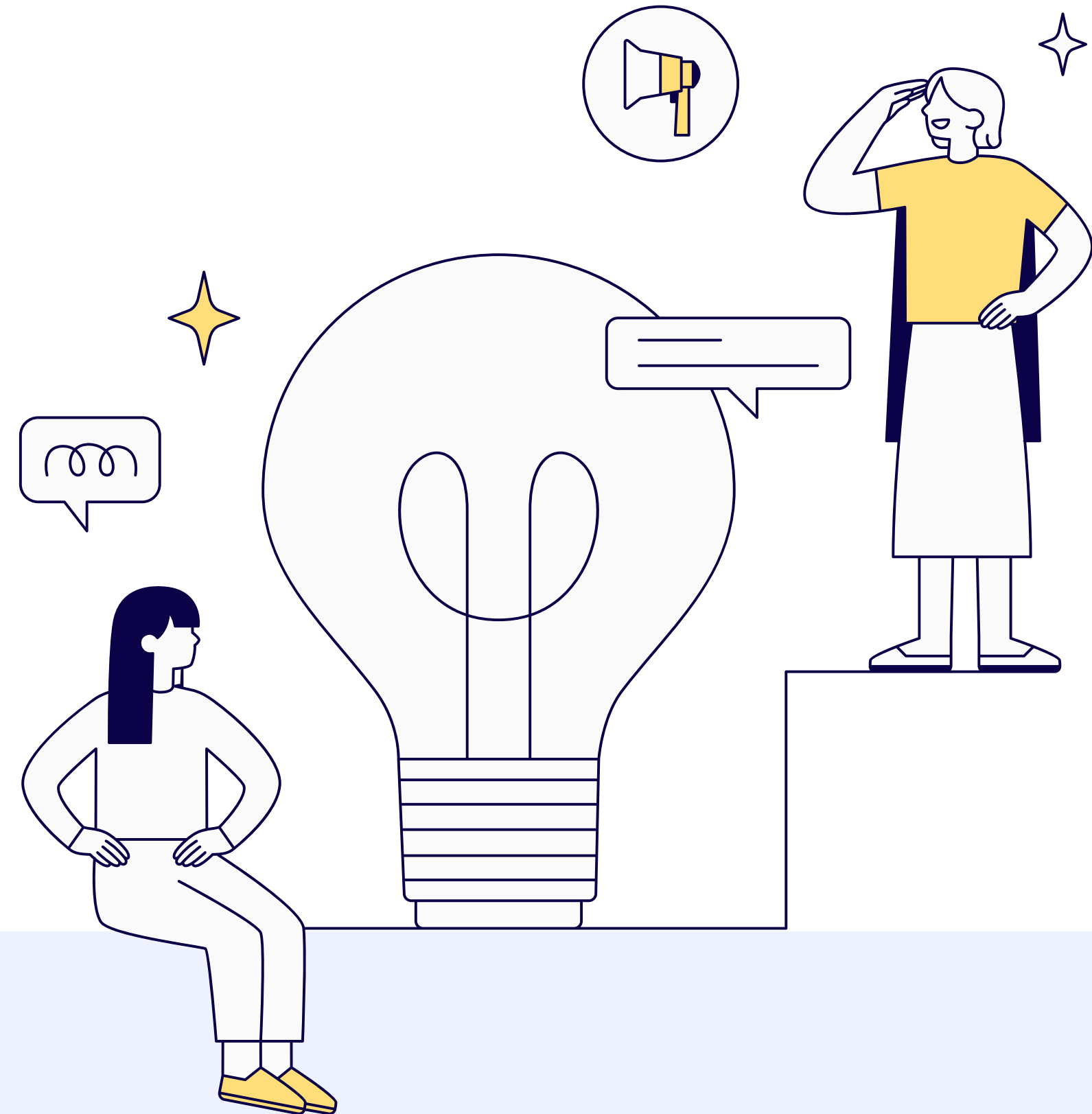
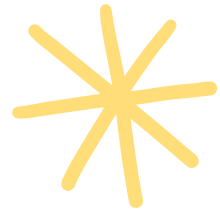
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# Introduction

## Understanding Ambiguity

A **CFG** is ambiguous if at least one string has more than one parse tree or derivation.



# Formal Defination

$G=(V,T,P,S)$ Where:

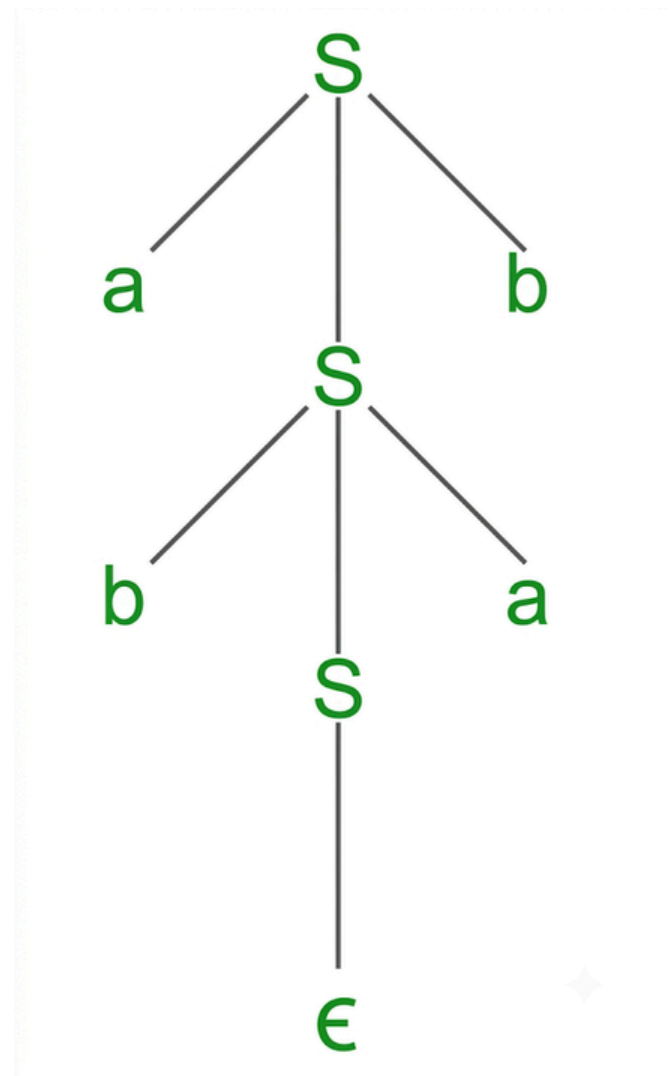
- V = Variables (Non-terminals)
- T = Terminals
- P = Production Rules
- S = Start Symbol



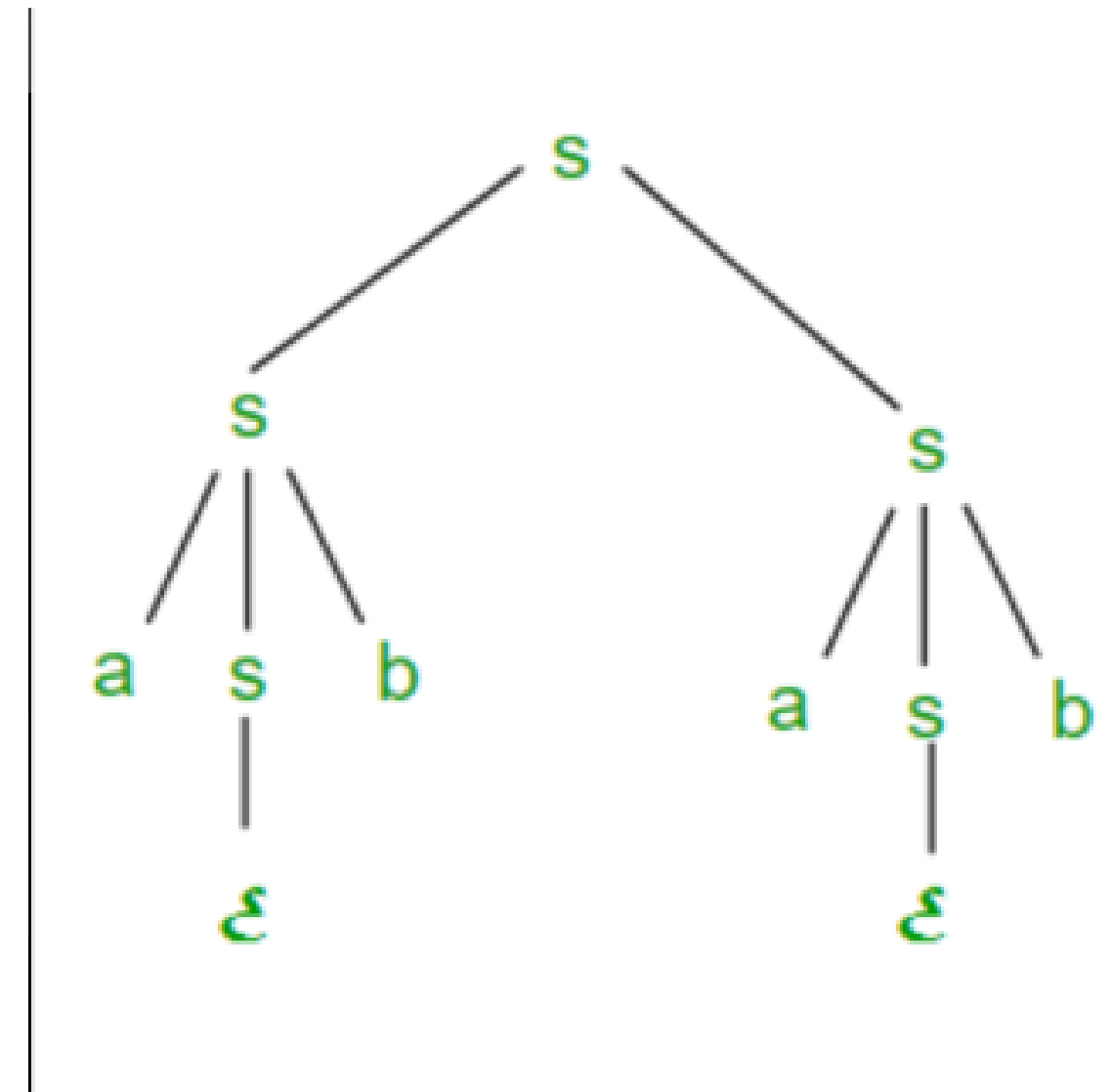
# Example

$$S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$$

$$S \rightarrow aSb \rightarrow abSab \rightarrow abab$$



$$S \rightarrow SS \rightarrow SaSb \rightarrow Sab \rightarrow aSbab \rightarrow abab$$



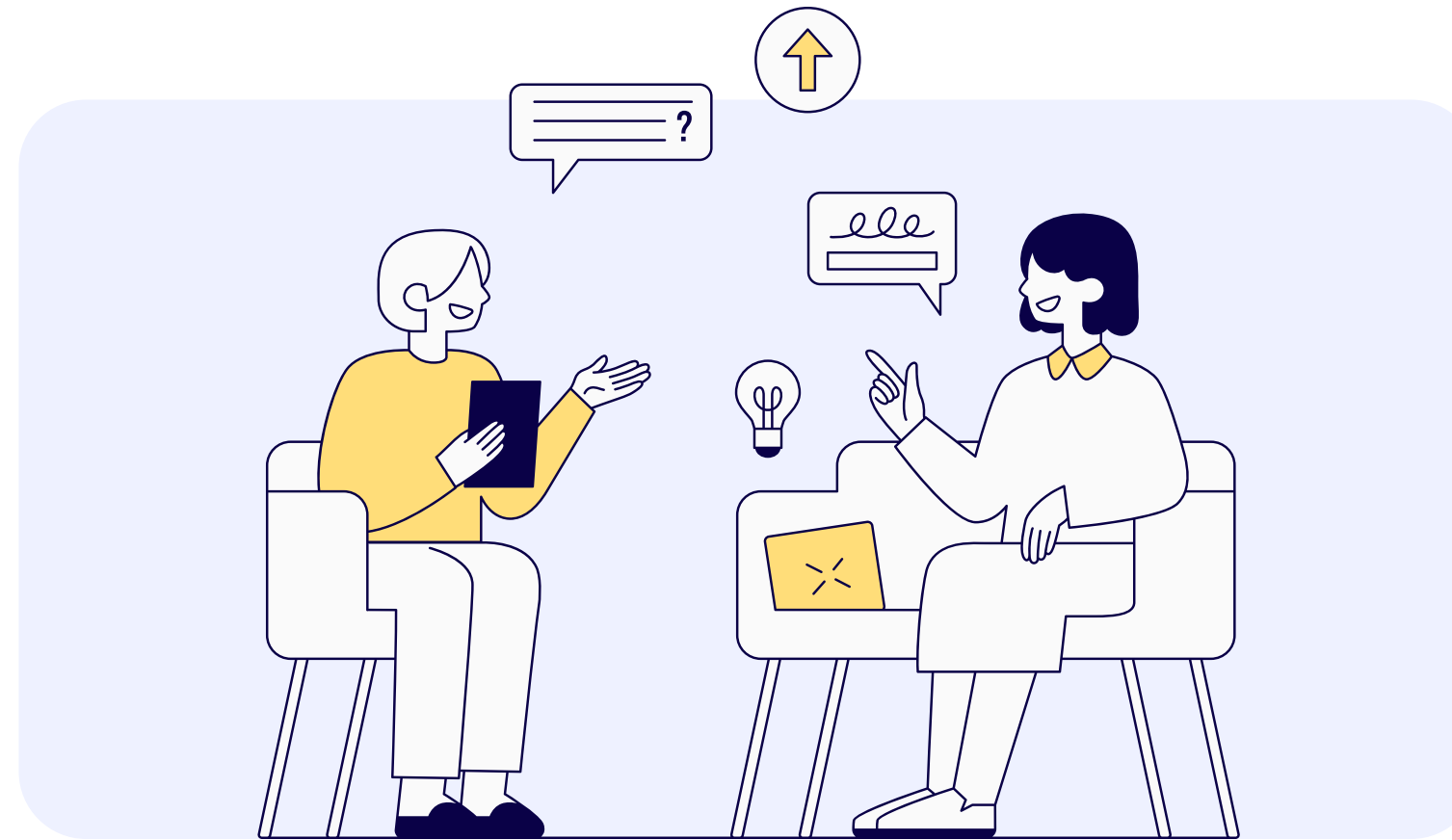


# Real-life example

## The Arithmetic Challenge

In languages like C, Java, or Python, expressions such as  $a + b * c$  are common. However, without strict rules, a computer could interpret this simple line in multiple conflicting ways. Interpretations must be 100% deterministic. Compilers cannot "guess" intent. Precision is the core of software stability.





### 1. Precedence First

The multiplication is performed first, then the addition.

Structure:  $a + (b * c)$

Matches standard math rules used by most languages.

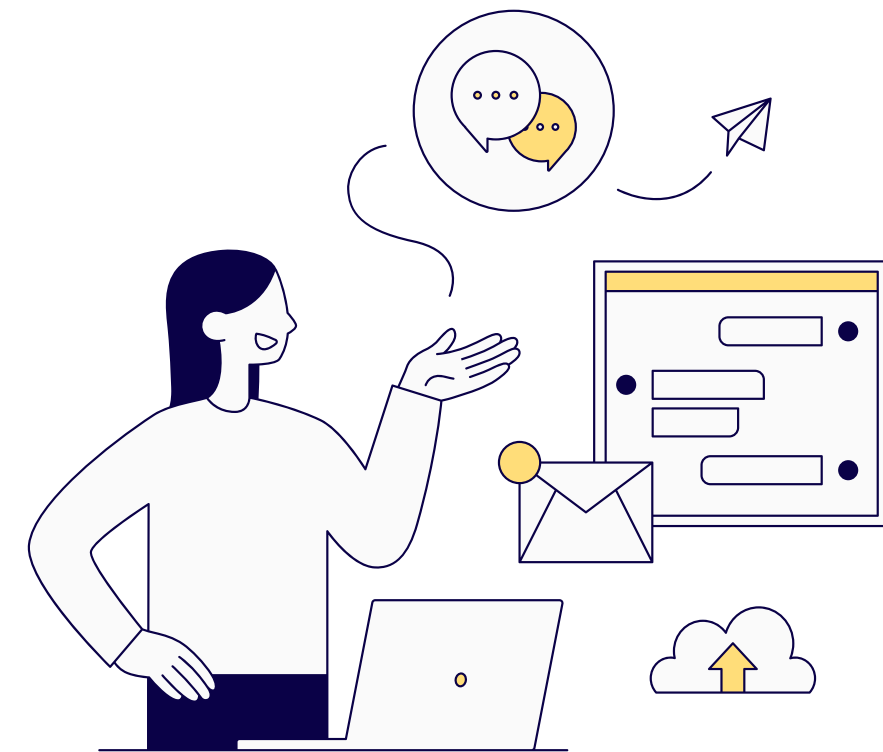
### 2. Sequence First

The addition is performed first, then the multiplication.

Structure:  $(a + b) * c$

Only valid if explicitly stated or grammar is flawed.

# Divergent Mathematical Outcomes



14  
Correct Outcome  
 $2 + (3 * 4)$

20  
Erroneous Outcome  
 $(2 + 3) * 4$

A single ambiguous grammar rule results in two distinct mathematical values for the same input.

# Conclusion: Ambiguity in CFG

Ambiguity occurs when one string can be interpreted in more than one way

This leads to multiple parse trees or derivations

In real-world programming:

The same expression can produce different results

This creates errors and confusion in programs

Therefore, ambiguity is not acceptable in programming languages

To solve this problem:

We design unambiguous grammars

We use operator precedence and clear structure



